

SMART INDIA HACKATHON 2026

ARES — Autonomous Rescue & Environmental Intelligence System



SMART INDIA
HACKATHON
2026

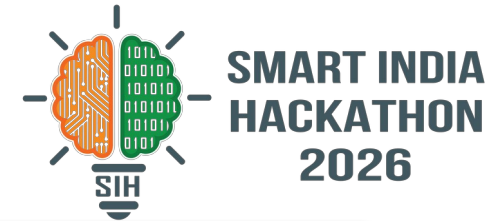
- Problem Statement ID - [fill from SIH portal]
- Problem Statement Title - [paste the exact PS title]
- Theme - [e.g. Disaster Management]
- PS Category - Hardware
- Team ID - [fill from SIH portal]
- Team Name - [as registered on portal]



3-minute demo video: [paste the public link here before exporting to PDF]

Your
Team
Name

A DRONE THAT DECIDES WHERE TO LOOK NEXT



THE PROBLEM

- A fixed grid gives every square metre the same attention — including empty ground.
- The battery runs out before the area is covered, and the survivor list arrives with no order of urgency.

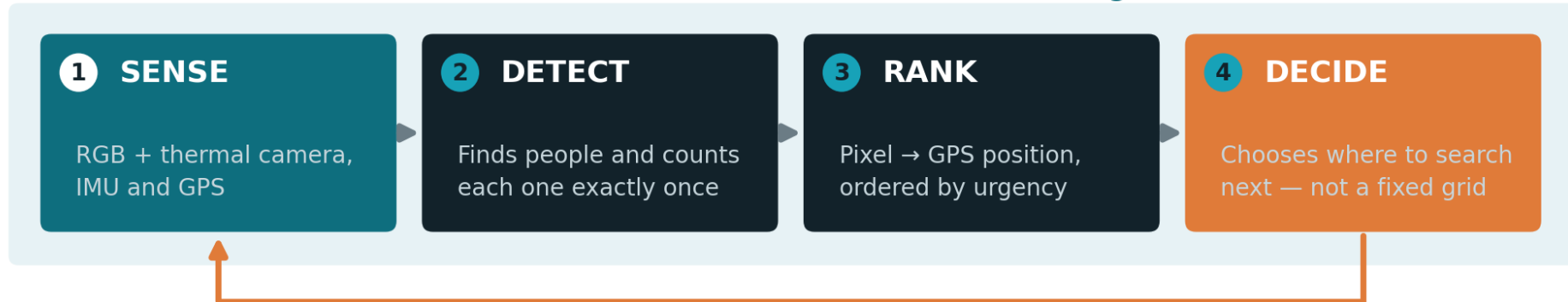
OUR SOLUTION

- While it flies, the drone builds a live probability map of where people are likely to be.
- It re-plans in the air and always flies to the best ground it has not yet searched.

WHY IT IS DIFFERENT

- Priority-driven search that never performs worse than a fixed grid.
- Detection, tracking and localization all run on the drone itself — no cloud, no mobile tower.

ALL FOUR STAGES RUN ON THE AIRCRAFT · no network, no cloud, no ground link



CLOSED LOOP · what the drone sees changes where it flies next — the ground station receives results, it does not make decisions

20 / 20

survivors found vs 11 / 20 for a fixed grid

3.2×

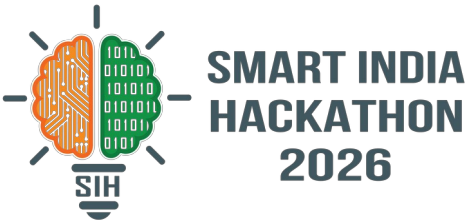
faster to reach half the survivors

100 %

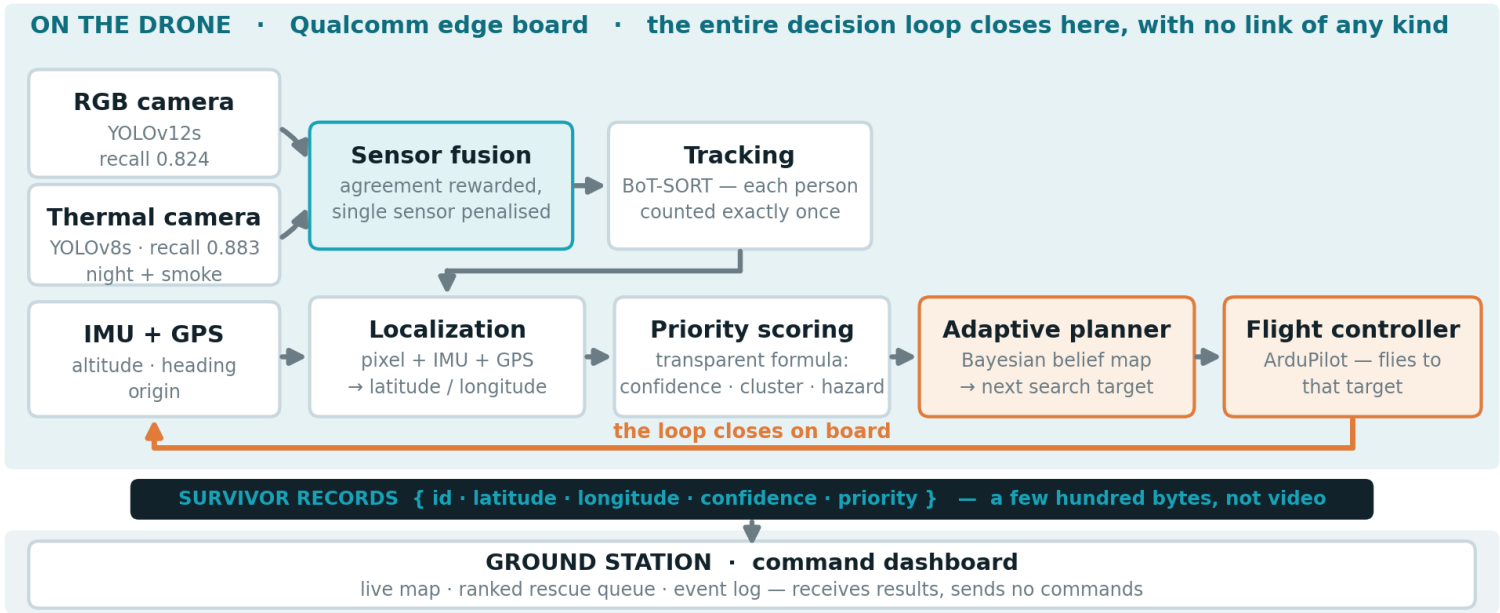
of model layers run on the drone's NPU

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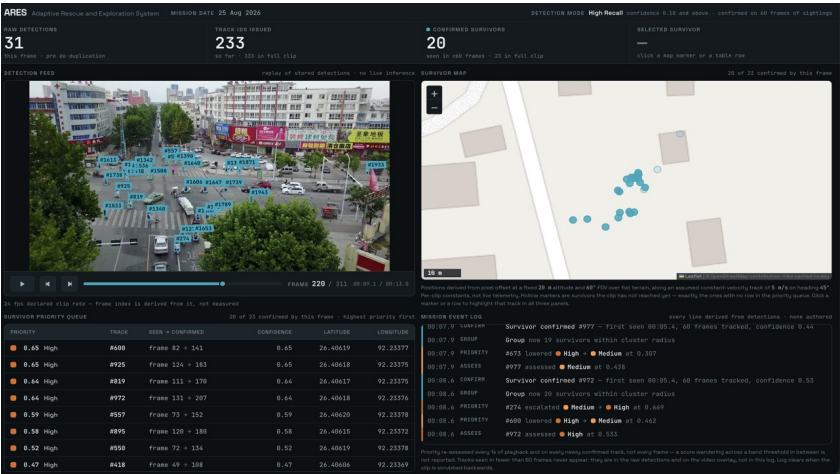
TECHNICAL APPROACH



YOLOv12s + YOLOv8s · BoT-SORT tracking · ONNX → INT8 on the Hexagon NPU · FastAPI + Pydantic · React + Leaflet · ArduPilot / MAVLink



THE COMMAND DASHBOARD — RUNNING, NOT A MOCKUP

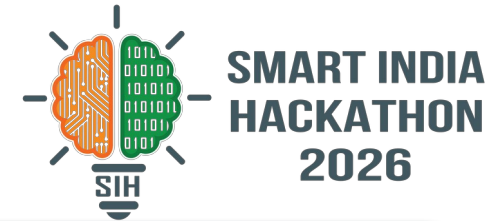


- Every count reconciles across all four panels
- Keeps running with the backend switched off

Everything that decides anything runs on the aircraft.
The ground station is where people look — not where choices are made.

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FEASIBILITY AND VIABILITY



BUILT AND MEASURED TODAY

Survivor detection

YOLOv12s · recall 0.824 over 86,092 labelled people

Thermal detection

YOLOv8s · recall 0.883 · works at night and in smoke

Runs on real edge silicon

209 ms/frame · 489 of 489 layers on the Hexagon NPU

Counts each person once

7,081 detections → 333 tracks → 23 confirmed survivors

Adaptive search

100 simulated missions · 20/20 found · 3.2× faster

Command dashboard

live map, ranked queue, event log · runs fully offline

RISKS, AND HOW WE HANDLE THEM

- Venue has no internet → map tiles cached locally; dashboard runs from a file
- Live inference could fail on stage → the demo replays genuine pre-computed model output
- No public dataset pairs aerial RGB with thermal → we measure fusion on the LLVIP benchmark and say so
- Small edge compute budget → YOLOv8n fallback already trained and benchmarked

IN PROGRESS — NUMBERS TO ADD BEFORE SUBMISSION

- Hazard classifier (fire/smoke, flood, collapsed structure): [per-class precision & recall — pending]
- RGB + thermal fusion on LLVIP: [false positives at equal recall — pending]
- INT8 accuracy after quantisation: [pending] · Closed-loop flight sim: [localization error — pending]

SCOPE — WHAT IS BUILT AND WHAT IS DESCRIBED

Built · measured

On-device inference · Priority alerting · Command dashboard · Geo-tagged mapping · Offline resilience

Partial · honest

Multi-sensor fusion (RGB + thermal + IMU + GPS) · Hazard classification, 3 classes of 7

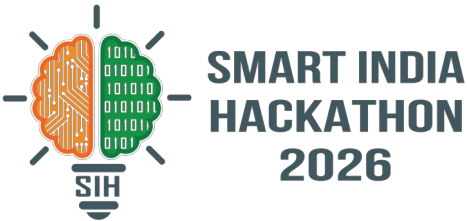
Simulated · not flown

Adaptive search planning · Autonomous navigation — flown on a simulated flight stack, never on hardware

We claim only what we measured. Nothing on this slide is estimated.

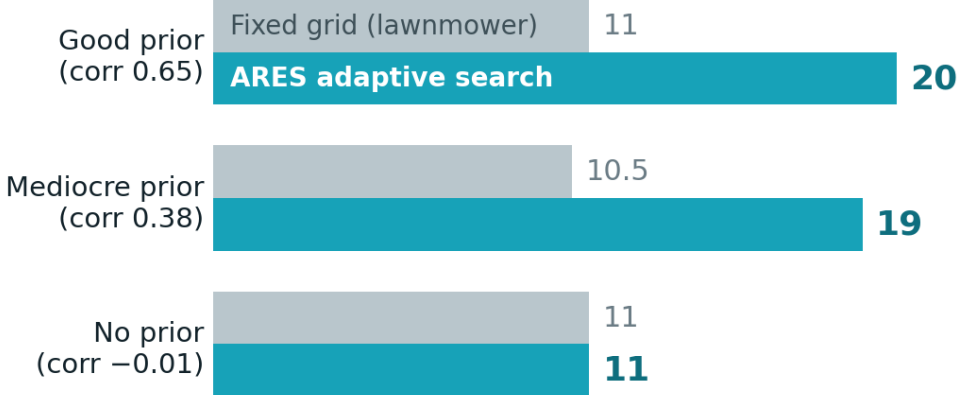
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IMPACT AND BENEFITS

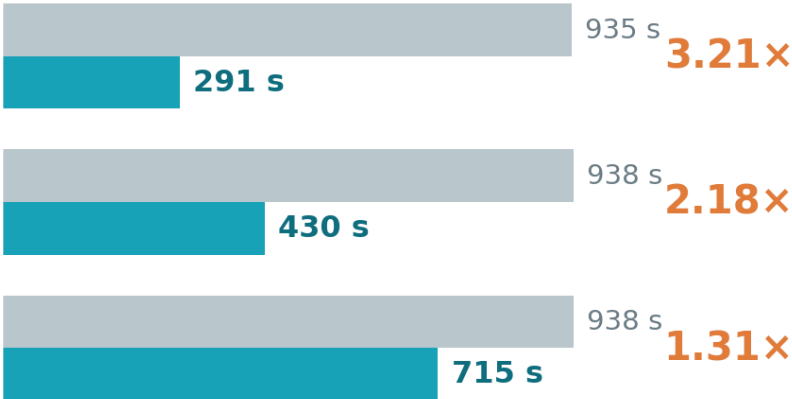


speed-up

Survivors found (out of 20)



Time to reach half the survivors



Median of 100 simulated missions · identical worlds, identical battery, identical detection probability (0.824, measured on 86,092 labelled people)

THE PRIOR — WHAT THE DRONE BELIEVES BEFORE IT FLIES

A map of where people are most likely to be — the drone's starting belief. The three rows above are three qualities of it: good, mediocre, and none at all.

IN THE FIELD IT IS ASSEMBLED FROM

- OpenStreetMap — buildings, shelters, schools
- Elevation model — high ground in a flood
- Sentinel-1 radar — flood extent, through cloud
- Emergency calls and last-known positions

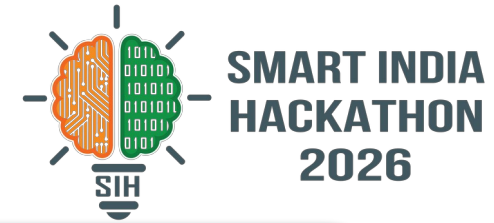
Quality is reported as correlation with the truth — 0.65, 0.38 and -0.01 above — never as a number we chose. Today the prior is synthetic; sourcing it from live data is the next step.

IMPACT

- Survivors reached sooner inside the 72-hour window
- A ranked rescue queue, not an unsorted list of dots
- Works with no network, no cloud and no mobile tower
- A commodity drone plus one edge board — affordable for a state response force

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RESEARCH AND REFERENCES



DATASETS USED

- C2A — casualty detection in disaster scenes
- VisDrone — UAV person detection benchmark
- HIT-UAV — real UAV thermal infrared imagery
- AResQ — aerial thermal rescue dataset (Sci Data, 2026)
- AIDER — UAV disaster-scene classification (hazards)
- LLVIP — registered visible/infrared pairs, for fusion
- COCO — pre-training baseline for both detectors
- OpenStreetMap — offline base map tiles for the dashboard

MODELS, TOOLS, HARDWARE

- Ultralytics YOLOv12s / YOLOv8s — detection + tracking
- BoT-SORT — multi-object tracking with re-identification
- Qualcomm AI Hub — Dragonwing RB3 Gen 2, IQ-9075 EVK
- ONNX + QNN — INT8 quantisation for the Hexagon NPU
- ArduPilot / MAVLink — flight stack and waypoints
- FastAPI · Pydantic · React · Leaflet · Vite
- PyTorch + ONNX Runtime — training and export
- Docker — reproducible flight-simulation environment

METHOD AND BACKGROUND

- Bayesian search theory — belief update over a grid
- Boustrophedon (lawnmower) coverage — our baseline
- Informative-path planning for UAV search and rescue
- NDRF / NDMA search-and-rescue operating guidance
- DGCA nano-category UAV rules (sub-250 g operation)
- Bayesian belief update with a staleness term — our coverage guarantee: skipped cells resurface later
- Tracking metrics: ID switches and fragmentation
- [add your faculty mentor's suggested references]

HOW OUR NUMBERS WERE PRODUCED

Accuracy is measured on a held-out split of 2,591 images and 86,092 labelled people. Latency is measured on real Qualcomm silicon through AI Hub — medians of ~100 runs, not best-case. Search results are medians of 100 simulated missions with identical worlds and identical battery. Demo footage is from public UAV datasets; every detection shown is genuine model output.

Demo video: [[paste the public link here before exporting to PDF](#)]